

Q1) EC2 Snapshot and AMI Creation

Launch an Ubuntu EC2 instance named **Lab_Q1_Apache** and configure the security group to allow SSH (Port 22) and HTTP (Port 80) access. Install the Apache2 web server and create a web page at **/var/www/html/index.html** displaying your **PRN number**. Verify that the website is accessible through the instance's public IP address.

Create an **EBS Snapshot** of the root volume attached to the instance and provide an appropriate description. Using the created snapshot, create an **AMI** named **Lab_Q1_Apache_snapshot**.

Launch a new EC2 instance named **snap_Lq1** using the AMI and ensure that the new instance is launched in a **different Availability Zone** from the original instance. Verify that the Apache website is accessible from the public IP address of the new instance and that it displays the same PRN number as the original instance.

Step 1: Launch Original EC2 Instance

UI Steps

1. AWS Console → **EC2**
2. Click **Launch Instance**

Configuration

Field	Value
Name	Lab_Q1_Apache
AMI	Ubuntu Server
Instance Type	t2.micro
Key Pair	Existing/New
Network	Default VPC
Auto Assign Public IP	Enable
Security Group	Allow HTTP (80) and SSH (22)

3. Click **Launch Instance**

 Screenshot: Original EC2 Created

Step 2: Connect and Install Apache

Connect

```
ssh -i key.pem ubuntu@<PUBLIC_IP>
```

Install Apache

```
sudo apt update
```

```
sudo apt install apache2 -y
```

```
ubuntu@ip-172-31-35-213:~$ sudo apt update

sudo apt install apache2 -y
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu
```

Verify Apache

```
systemctl status apache2
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-35-213:~$ systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-06-07 20:23:49 UTC; 17s ago
 Invocation: 191b4a401eac4594b15945a03814fa9e
    Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 2368 (apache2)
   Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec:"
     Tasks: 55 (limit: 627)
   Memory: 5.6M (peak: 5.9M)
      CPU: 89ms
   CGroup: /system.slice/apache2.service
           └─2368 /usr/sbin/apache2 -k start -DFOREGROUND
             └─2384 /usr/sbin/apache2 -k start -DFOREGROUND
               └─2385 /usr/sbin/apache2 -k start -DFOREGROUND
```

Step 3: Create Web Page

Command

```
echo "<h1>PRN: 260240123060</h1>" | sudo tee /var/www/html/index.html
```

Verify

```
cat /var/www/html/index.html
```

 Screenshot: index.html Content

```
ubuntu@ip-172-31-35-213:~$ cat /var/www/html/index.html
<h1>PRN: 260240123060</h1>
```

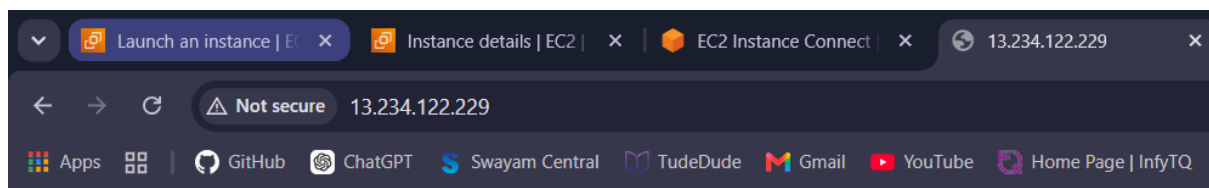
Step 4: Verify Website

Browser

http://<PUBLIC_IP>

Expected:

PRN: 260240123060



PRN: 260240123060

 Screenshot: Website Working

Step 5: Create EBS Snapshot

UI Steps

1. EC2 → Instances
2. Select **Lab_Q1_Apache**
3. Storage Tab
4. Click Volume ID

Create Snapshot

1. Select Volume

2. Actions → Create Snapshot

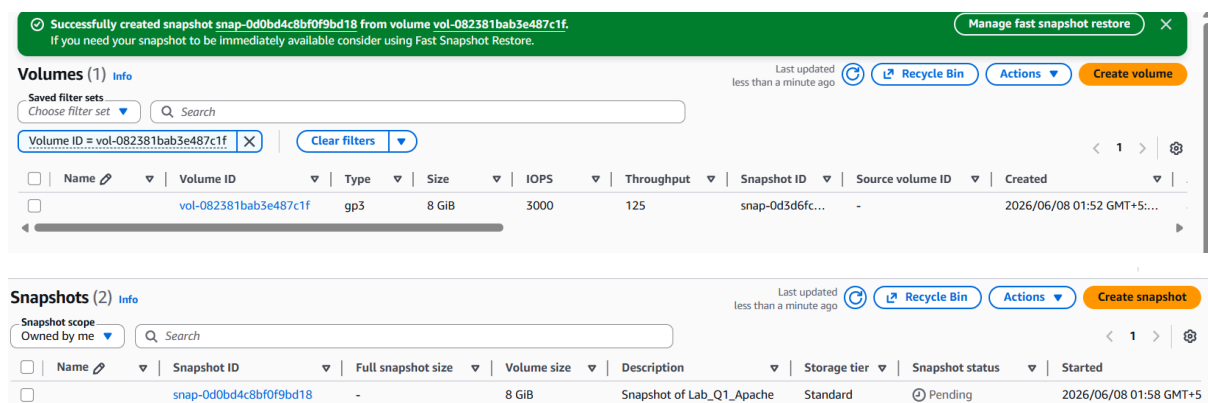
Configuration

Field	Value
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Description	Snapshot of Lab_Q1_Apache
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3. Click **Create Snapshot**

 Screenshot: Snapshot Created



Successfully created snapshot `snap-0d0bd4c8bf0f9bd18` from volume `vol-082381bab3e487c1f`. If you need your snapshot to be immediately available consider using Fast Snapshot Restore.

Manage fast snapshot restore

Volumes (1) Info

Saved filter sets: Choose filter set

Volume ID: `vol-082381bab3e487c1f` Clear filters

	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source volume ID	Created
☐		<code>vol-082381bab3e487c1f</code>	gp3	8 GiB	3000	125	<code>snap-0d3d6fc...</code>	-	2026/06/08 01:52 GMT+5...

Snapshots (2) Info

Snapshot scope: Owned by me

	Name	Snapshot ID	Full snapshot size	Volume size	Description	Storage tier	Snapshot status	Started
☐		<code>snap-0d0bd4c8bf0f9bd18</code>	-	8 GiB	Snapshot of Lab_Q1_Apache	Standard	Pending	2026/06/08 01:58 GMT+5

Step 6: Create AMI

UI Steps

1. EC2 → AMIs
2. Click **Create AMI**

OR

1. Select Snapshot
2. Actions → Create Image

Configuration

Field	Value
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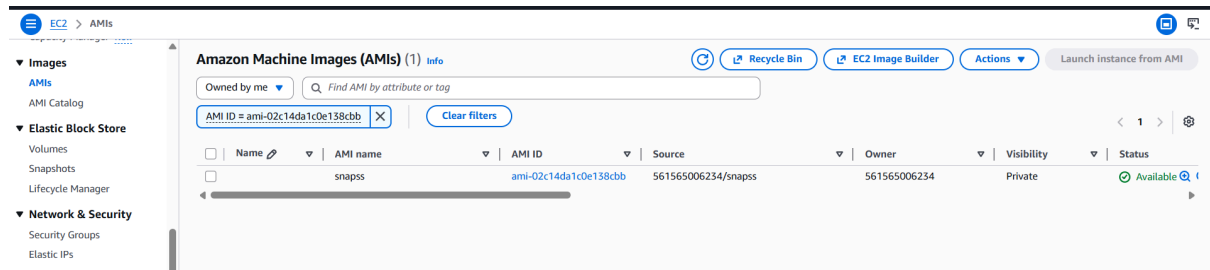
Name	Lab_Q1_Apache_snapshot
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3.
Click **Create AMI**

Wait until:

Available

 Screenshot: AMI Available



Step 7: Launch New EC2 from AMI

UI Steps

1. EC2 → AMIs
2. Select: Lab_Q1_Apache_snapshot
3. Click **Launch Instance from AMI**

Configuration

Field	Value
Name	snap_Lq1
Type	t2.micro
Key Pair	Same Key
Security Group	HTTP + SSH

Step 8: Launch in Different Availability Zone

UI Steps

Network Settings → Edit

Select a different AZ.

Example:

Original Instance	New Instance
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ap-south-1a

ap-south-1b

Launch Instance.

 Screenshot: New Instance Created

☐	snap-Lq1	i-003ec46ff67b7d4d2	Running	t3.micro	Initializing	ap-south-1b	ec2-15-207-113-26.ap-...	15.207.113.26	-
☐	Lab_Q1_Apache	i-0dd6c8460b4a64be	Running	t3.micro	3/3 checks passed	ap-south-1a	ec2-13-234-122-229.ap...	13.234.122.229	-

Step 9: Verify Website on New Instance

Copy new Public IP.

Open browser:

http://<NEW_PUBLIC_IP>

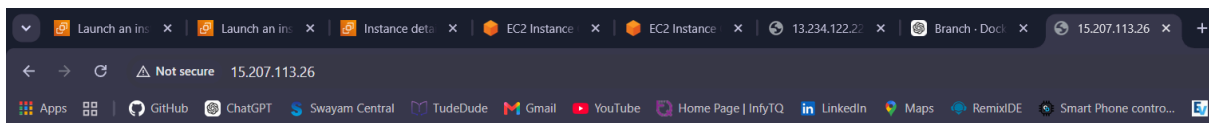
Expected:

PRN: 260240123060

Same page should appear.

 Screenshot: Website on New Instance

```
root@ip-172-31-14-172:~# curl localhost
<h1>PRN: 260240123060</h1>
root@ip-172-31-14-172:~#
```



PRN: 260240123060

Step 10: Display Resource Details

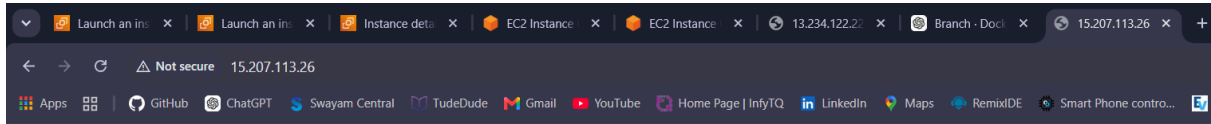
Original EC2

EC2 → Instances

Show:

Lab_Q1_Apache

 Screenshot



PRN: 260240123060